TEST CODE **000592**

FORM TP 9888

MAY/JUNE 1998

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

MATHEMATICS

Paper 02 - General Proficiency

2 hours 40 minutes

04 JUNE 1998 (a.m.)

INSTRUCTIONS TO CANDIDATES

1. **Answer ALL questions in Section I, and any TWO in Section II.**
2. **Begin the answer for each question on a new page.**
3. **Full marks may not be awarded unless full working or explanation is shown with the answer.**
4. **Mathematical tables, formulae and graph paper are provided.**
5. **Silent electronic calculators may be used for this paper.**
6. **You are advised to use the first 10 minutes of the examination time to read through this paper. Writing may begin during this 10-minute period.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

Copyright © 1997 Caribbean Examinations Council.
All rights reserved.

SECTION I

Answer ALL the questions in this section.

ALL working must be clearly shown.

1. (a) Calculate the exact value of

$$\frac{2\frac{1}{7} - \frac{1}{3}}{1\frac{6}{7}}$$

(2 marks)

(b) Express $\frac{0.0402}{0.71}$

(i) correct to 2 decimal places

(ii) correct to 2 significant figures

(iii) in standard form. **(3 marks)**

(c) (i) The basic wage rate for a technician for a 42-hour week is $36.40 per hour. Calculate the technician's weekly wage.

(ii) For overtime work the technician is paid one and a half-times the basic hourly rate. Calculate the technician's total wage for 58 hours of work.

(iii) In a new wages agreement, the basic rate of payment is $1490.00 for 36 hours of work. Calculate:

– the technician's hourly rate of pay

– the percentage increase in the hourly rate. **(7 marks)**

Total 12 marks

2. (a) Factorize completely:

(i) $2y^2 + 3y$

(ii) $81 - m^2$

(iii) $2x^2 - x - 15$ **(4 marks)**

(b) If S varies directly as $(r + 1)$, and $S = 8$ when $r = 3$, calculate the value for r when $S = 50$. **(4 marks)**

GO ON TO THE NEXT PAGE

(c) Ninety tickets were sold for a concert. x tickets were sold for \$3.00 each and the rest of the tickets were sold for \$4.00 each.

Write an expression in x to represent the number of tickets sold at

(i) \$3.00 each

(ii) \$4.00 each.

(iii) If the total sales on all tickets amounted to \$300.00, how many of the tickets costing \$3.00 were sold? **(4 marks)**

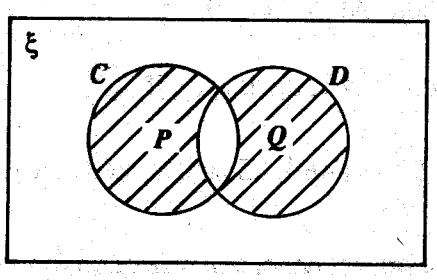
Total 12 marks

3. (a) The Universal set, ξ, contains the sets J, K and L such that

$$J \not\subset K,\ J \cap K \neq \phi \text{ and } L \subset (J \cup K)'.$$

Draw a Venn diagram to show the relation amongst the sets J, K and L. **(3 marks)**

(b) The Venn diagram below shows sets C and D, where P and Q are subsets of C and D respectively.



(i) Given that $n(C) = 33$, $n(D) = 41$, $n(C \cap D) = x$, write down, in terms of x, the values of $n(P)$ and $n(Q)$.

(ii) If $n(C \cup D) = 51$, calculate the value of x. **(3 marks)**

(c) The matrix $H = \begin{pmatrix} h & 2 \\ 2 & -h \end{pmatrix}$.

(i) Determine H^2.

(ii) Evaluate h, if $H^2 = 5\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. **(4 marks)**

Total 10 marks

GO ON TO THE NEXT PAGE

4. (a) (i) All construction lines must be clearly shown.

Draw a line PR which measures 7.5 cm. Construct the perpendicular bisector, QS, of PR to meet PR at O, such that OQ and OS both measure 5 cm. Complete the parallelogram $PQRS$.

(ii) State the name of this type of parallelogram $PQRS$.

(iii) Measure and state, in cm, the length of PQ.

(iv) Measure and state the size of angle QPS. **(6 marks)**

(b) On the graph paper provided for this question, the point $P(x, y)$ has been marked in.

(i) Write down the coordinates of P.

(ii) Through P, draw a straight line whose y-intercept is 4.

(iii) Calculate the gradient of the straight line.

(iv) Determine the equation of the straight line. **(6 marks)**

Total 12 marks

5. (a) $\overline{AB} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ and $EF = \begin{pmatrix} 5 \\ e \end{pmatrix}$

(i) Calculate $|\overline{AB}|$, the length of $\overline{AB}$.

(ii) If $|\overline{EF}| = 3|\overline{AB}|$, show that the value of $e = 10\sqrt{2}$. **(4 marks)**

(b) The transformation, M, denotes a reflection in the line $x = 2$. The transformation, T, denotes a translation represented by $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$.

(i) On graph paper, using a scale of 2 cm to 1 unit on both the x and y axes, plot the points A (4, 2) and B (3, 1). Join AB.

On your graph:

(ii) Draw the image $A'B'$ of AB under the transformation, M.

(iii) Draw the image $A''B''$ of AB under the transformation, T.

(iv) Describe, geometrically, a single transformation which will map $A''B''$ onto $A'B'$. **(8 marks)**

Total 12 marks

GO ON TO THE NEXT PAGE

6. (a) Given $f(x) = x^2$ and $g(x) = 5x + 3$,

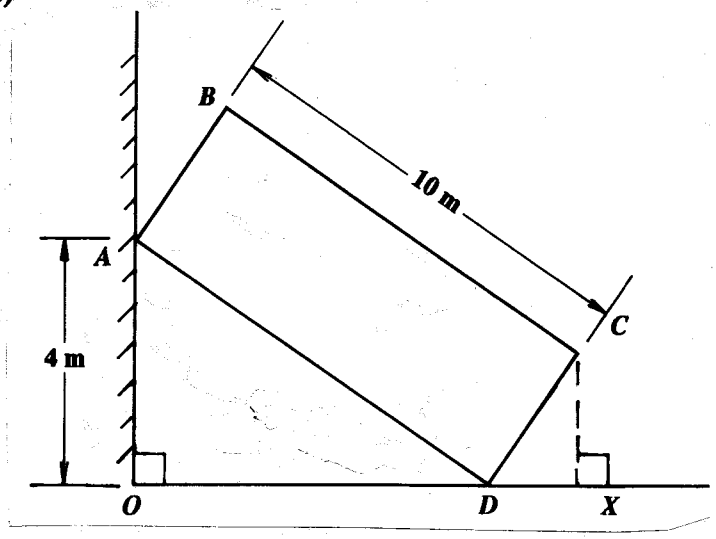
calculate:

(i) $f(-2)$

(ii) $gf(-2)$

(iii) $g^{-1}(x)$ **(7 marks)**

(b)



The diagram above shows a rectangular sheet of metal, *ABCD*, supported by a vertical wall (shaded), which is at right angles to the level ground *OX*. *AB* measures 3 metres and *AD* measures 10 metres. *A* is 4 metres above *O*.

(i) Calculate the size of the angle *ODA*.

(ii) Hence, calculate the size of the angle *CDX*.

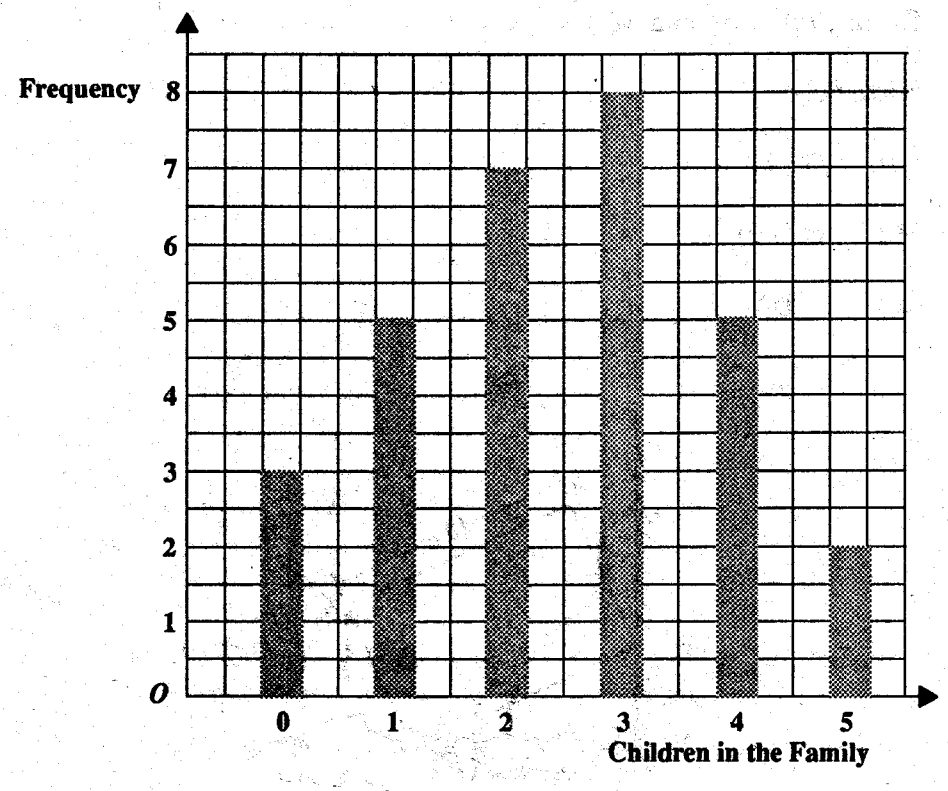
(iii) If *CX* represents the height in metres of *C* above the ground, calculate *CX*.

(4 marks)

Total 11 marks

GO ON TO THE NEXT PAGE

7.



The graph above shows the number of children in 30 randomly chosen families.

(a) Using the graph construct a frequency distribution table to represent the data for the children of the 30 families. **(2 marks)**

(b) Determine the mean and the median of the distribution. **(5 marks)**

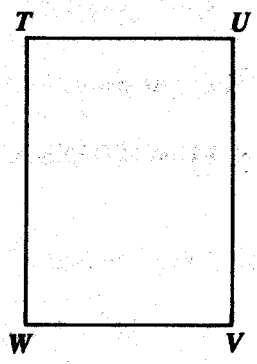
(c) Calculate the probability that a child, chosen at random, comes from a family with at least 4 children. **(2 marks)**

(d) If the sample size were increased to 42 families, how many families would be expected to have at least 4 children? **(2 marks)**

Total 11 marks

GO ON TO THE NEXT PAGE

8.



For this question, take π to be $\frac{22}{7}$.

The rectangle, *TUVW*, represents a sheet of metal, whose length, *UV*, is 100 cm and its breadth, *WV*, is 88 cm. The sheet of metal is formed into an open cylinder, by bringing *T* onto *U* and *V* onto *W*.

(a) Sketch a diagram of the cylinder, and clearly mark in 100 cm as a dimension of the cylinder. **(1 mark)**

(b) Calculate the radius, in cm, of the base of the cylinder. **(2 marks)**

(c) Calculate the area of the base of the cylinder. **(2 marks)**

(d) If the cylinder is closed at one end, calculate, in cm^3, the capacity of the cylinder. **(2 marks)**

(e) If 50 litres of liquid is poured into the cylinder, calculate, in cm, the depth of the liquid in the cylinder. **(3 marks)**

Total 10 marks

SECTION II

Answer TWO questions in this section.

RELATIONS, FUNCTIONS AND GRAPHS

9. (a) Calculate the set of values of x for which $x^2 + 6x - 5 > 2$. **(3 marks)**

(b) (i) Given that $f(x) = 2x^2 + 2x - 1$, copy and complete the table below.

x	-3	-2	-1	0	1	2	3
$f(x)$		3		-1		11	23

(ii) Using 1 cm to represent 1 unit on the x-axis and 1 cm to represent 5 units on the $f(x)$ axis, draw the graph of $f(x) = 2x^2 + 2x - 1$ for $-3 \leq x \leq 2$.

(iii) On the graph for $f(x) = 2x^2 + 2x - 1$, draw in the axis of symmetry for $f(x)$, and state the value of x where this occurs.

(iv) From your graph, obtain estimates for:

- the values of x for which $f(x) = 0$
- the set of values of x for which $f(x) < 0$
- the least value which $f(x)$ can take. **(12 marks)**

Total 15 marks

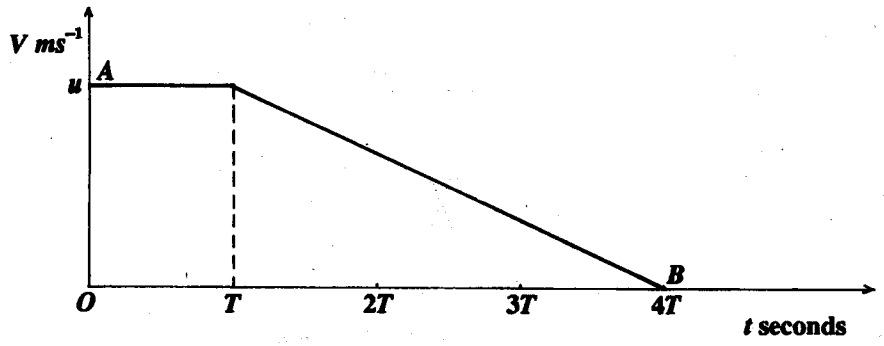
10. (a) A car starts from rest and accelerates for 5 seconds to a point, P, reaching a velocity of $30\ \text{ms}^{-1}$. It maintains this velocity for 10 seconds.

(i) Using 1 cm to represent 2 seconds on the time axis and 1 cm to represent 5 metres/sec. on the velocity axis, draw the velocity-time graph for this part of the car's journey.

(ii) Using your graph, calculate:

- the velocity of the car after 2 seconds
- the car's acceleration in the first 5 seconds
- the distance travelled in the 15 seconds
- the average speed for the 15 seconds. **(8 marks)**

(b)



The velocity-time graph, **not drawn to scale**, shows the movement of a particle along a straight line. The particle passes a point, A, when $t = 0$. It maintains a constant speed, $u ms^{-1}$, for T seconds, and then decelerates uniformly before coming to rest at B.

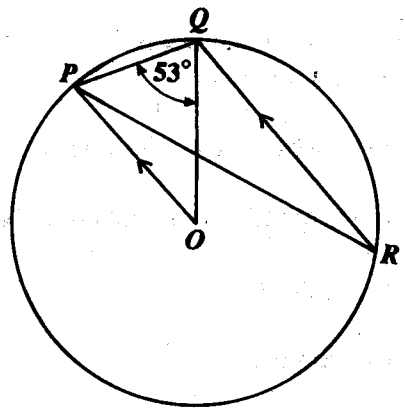
(i) If the average speed for the whole movement is $40\ ms^{-1}$, calculate the value of u.

(ii) Given that the deceleration between T seconds and $4T$ seconds is $5\ ms^{-2}$, calculate the value of T. **(7 marks)**

Total 15 marks

GEOMETRY AND TRIGONOMETRY

11. (a)



In the figure above, **not drawn to scale**, PQR is a circle with centre, O. The angle $PQO = 53°$. PO is parallel to QR.

Calculate, giving reasons to support your answers, the size, in degrees, of the angles:

(i) POQ

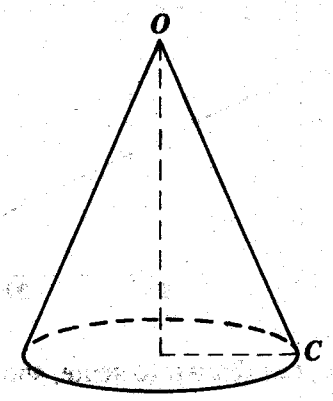
(ii) OQR

(iii) PRQ

(iv) QPR **(8 marks)**

GO ON TO THE NEXT PAGE

(b)



The diagram represents a right circular cone with a base radius of 9 cm and a slant height of 15 cm. The cone is cut along OC, and unrolled to form a sector, OCC', from a circle of centre, O. **[Take π to be 3.14]**

Calculate:

(i) The length, in cm, of the arc, CC'

(ii) The size, in radians, of the angle COC' subtended by the arc CC'

(iii) The area, in cm^2, of the curved surface of the cone **(7 marks)**

Total 15 marks

12. A cross-country race covers a course which takes the shape of a triangle, KLM. Runners are to start at K and run 4 km due north to a point, L.

From L, they are to run 5 km on a bearing of 300° to a point, M, and then run directly back to K.

(a) Draw a diagram to represent the triangular course carefully labelling:

(i) The points K, L and M

(ii) The distances 4 km and 5 km

(iii) The north direction

(iv) The bearing 300° **(5 marks)**

(b) Calculate:

(i) The distance KM in km

(ii) The bearing of M from K **(10 marks)**

Total 15 marks

GO ON TO THE NEXT PAGE

VECTORS AND MATRICES

13. (a) (i) Write down the inverse of the matrix $\begin{pmatrix} 1 & 4 \\ 2 & 1 \end{pmatrix}$.

(ii) Hence or otherwise determine the value for a, b, c and d in the matrix equation

$$\begin{pmatrix} 1 & 4 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 4 & 13 \\ 8 & 5 \end{pmatrix}.$$

(5 marks)

(b) The coordinates of the points, E, D and F, are $(1, 3)$, $(2, -1)$ and (m, n) respectively. $\overrightarrow{OE}$, $\overrightarrow{OD}$ and $\overrightarrow{OF}$ are position vectors relative to the origin, O.

(i) Express $\overrightarrow{OE}$, $\overrightarrow{OD}$ and $\overrightarrow{OF}$ in the form $\begin{pmatrix} x \\ y \end{pmatrix}$.

(ii) Given $\overrightarrow{OF} = 3\,\overrightarrow{OE} - \overrightarrow{OD}$, determine the values of m and n.

(iii) If G is the midpoint of OD, use a vector method to prove

– $EF : GE = 2 : 1$

– G, E and F are collinear.

(10 marks)

Total 15 marks

14. (a) If $\begin{pmatrix} 5 & 4 \\ m-1 & 2 \end{pmatrix}$ is a singular matrix, calculate the value of m. **(3 marks)**

(b) (i) The matrix $R = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$. Determine the coordinates of the image of $(1, 2)$ under the transformation R when $\theta = 90°$.

(ii) If the point $(p, 3)$ is on the line (L) given by $x + 2y = 5$, calculate the value of p.

(iii) Given that the point $(1, 2)$ is on L, determine the image L' of the line L under the transformation R.

(iv) Write the matrix equation to represent the pair of simultaneous equations given by L and L'. **(12 marks)**

Total 15 marks

END OF TEST

TEST CODE **000592**

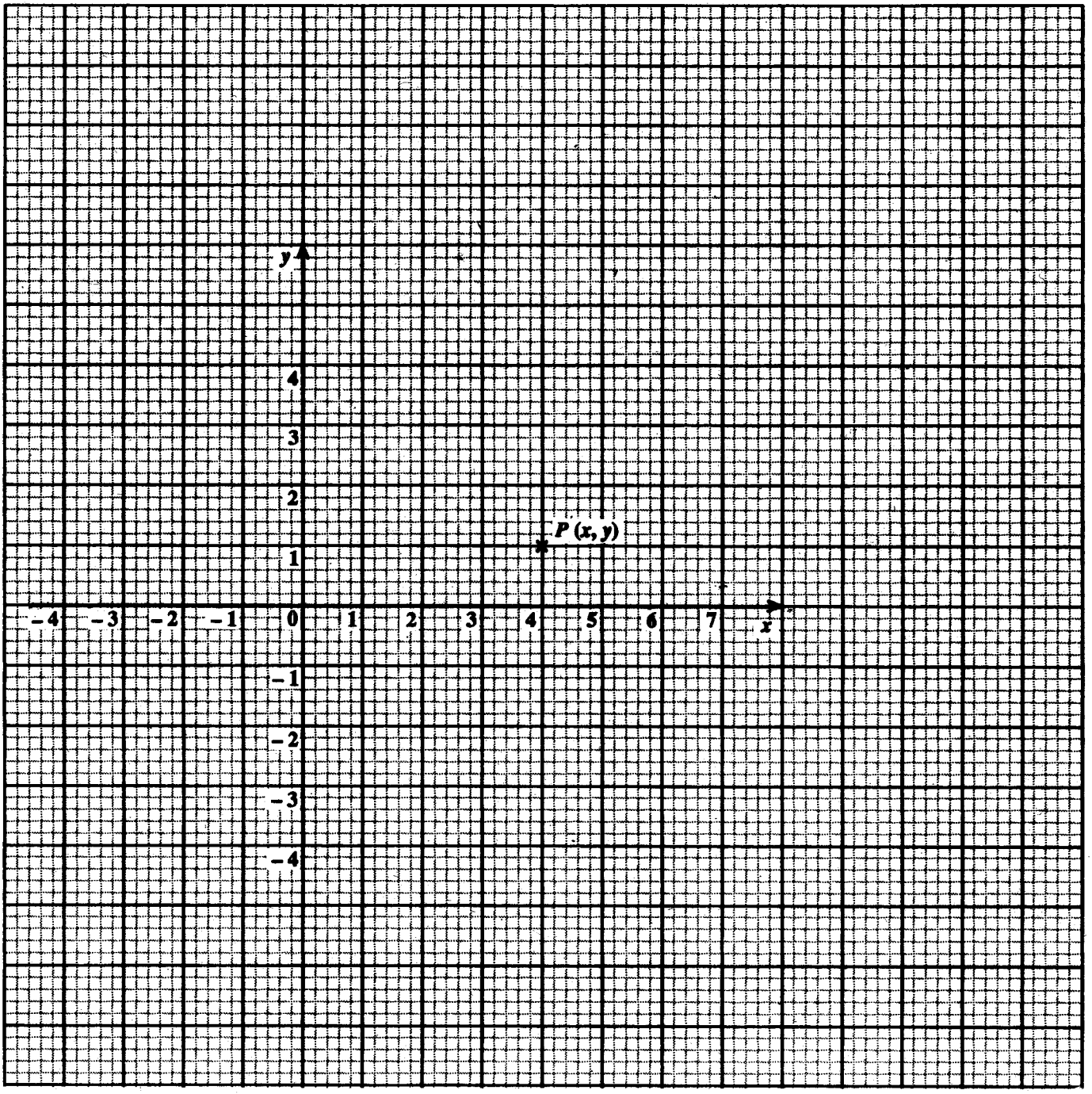
FORM TP 9888

MAY/JUNE 1998

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

MATHEMATICS

Paper 02 - General Proficiency

Answer sheet for Question 4 (b) **Candidate number .....**